Static and kinetic friction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish adjustable static friction from sliding friction.

SUCCESS CRITERIA - I can...

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “friction” and “static” to compare your model or result. Sentence starters are optional.

friction: a contact force opposing relative sliding or its tendency

static: not sliding relative to the contacting surface

kinetic: associated with motion; kinetic friction acts during sliding

01 Warm-up: bring back an earlier idea

A runner pushes the ground backward with 50 N.
Name the partner force, its size and the object it acts on.
Explain its role in the runner’s motion.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $0 \le f_s \le \mu_s N$; $f_k = \mu_k N$.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

Worked reasoning: A box remains still while pushed with 3 N. Static friction is 3 N opposite the push, provided its maximum is at least 3 N. Static friction need not equal its maximum.

Your turn: Draw the horizontal forces. Then redraw for a 5 N push that is still below the maximum.

Sentence starter: The … in my model represents … . Its label shows … .



Box remains at rest. Redraw for a 5 N push.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Static and kinetic friction

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Below the limit

The box stays at rest with an 8 N push right. Maximum static friction is 15 N.
Find actual friction and net force.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.



The dot represents the object. At rest; static limit 15 N

Marzano check - circle: ① ② ③

04 Crossing the limit

The push on the box in question 3 rises to 18 N. Once it slides right, kinetic friction is 12 N. Complete the diagram and find the net force.
Explain why you use 12 N rather than the 15 N static limit.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

05 Can friction help?

A runner's foot pushes backward against the ground without slipping.
Draw the friction force the ground exerts on the foot.
Explain why "friction always points backward" is not a reliable rule.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

Static and kinetic friction

antocciphysics

Continue down each column.

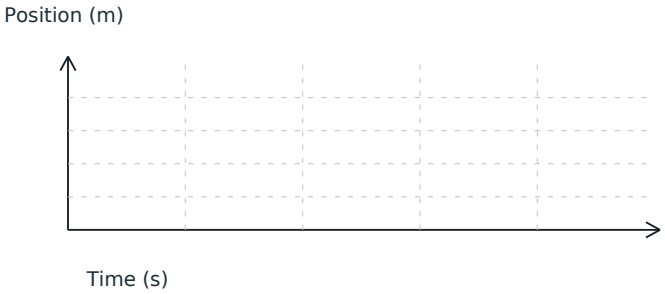
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A cart starts at 4 m and moves right at 6 m/s for 3 s.
Predict final position. Allowed endpoints are 20 to 24 m; state this interval and one assumption.
Label the model with the given positions and any prediction.

Word help: final = at the end of the interval; subscript f

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: 1 2 3

07 Exit check

A box has a maximum static friction of 20 N. In one trial, a 12 N push leaves it at rest. In a new trial, a 25 N push starts it sliding with 16 N kinetic friction.
Find actual friction in the first trial and net force in the second.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Trial	Push (N)	Friction (N)	Net force (N)
At rest	12	____	____
Sliding	25	16	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A box has maximum static friction 8 N and kinetic friction 6 N.
Predict the behavior for pushes of 4, 8 and 10 N.
Draw the sliding case.

Sentence starter: I predict … . My model supports this because … .



Add and label force arrows.

Draw the sliding case, with actual friction.

Marzano check - circle: 1 2 3